1.1-1.2

Domain x-axis
Range y-axis

[ ] included
( ) not-included

- We must scan the graph from left to right.
- Vertical line test: must only touch 1 line (if it passed then it is a function)
* A horizontal parabola is not a function of x because it fails the test.
- ○ & • have different meanings
- ± means 2 values which are not functions.
- Piecewise means there are many functions with different rules

Absolute value of $|x|$ in a piecewise → $|x| = \begin{cases} x & x>0 \\ -x & x<0 \end{cases}$

- Even when $f(-x) = f(x)$
- Odd when $f(-x) = -f(x)$
- The graph of even is like a card (flipped virtually). Odd is the opposite.

1.2

Linear function $y = mx + b$ ← b is the y-intercept $f(x) = 2x + 3$

$m = 2 \quad b = 3$

Slope form: $m = \frac{y_2 - y_1}{x_2 - x_1}$

The slope for horizontal is 0 & is undifined for vertical

point-slope form: $y - y_1 = m(x - x_1)$

There are 3 Stages of polynomial functions:

1st degree: There are only 2 ~~variables~~ variables
2nd degree: There are 3 variables
3rd degree: There are 4 variables

a poly fraction is called a rational function

power functions

odd :



even :



$f(x) = 3^{x-2} + 4$

↑ The sign effects the range if it starts with $-\infty$ or ∞

↑ Asymptote & Range

1.3 - 1.4

$f \circ g(x)$ → $f(g(x))$ $\quad$ $g \circ f(x)$ → $g(f(x))$ $\quad$ $f \circ g \circ h(x)$ → $f(g(h(x)))$

Decomposition: is breaking & Identifying the respective letter.

Graph transformation

(Shifting)

$y = f(x) + c$ (up) $\quad$ $y = f(x) - c$ (down)

$y = f(x + c)$ (left) $\quad$ $y = f(x - c)$ (Right)

(Reflecting)

$y = -f(x)$ (x-axis)

$y = f(-x)$ (y-axis)

1.4 - 1.5

Exponential function → $f(x) = b^x$

when it comes to graphs
- $0 < b < 1$: decrease
- $b = 1$: constant
- $b > 1$: increase

Domain : $(-\infty, \infty)$

Range if $b \neq 1$ $(0, \infty)$, $b = 1$ $\{1\}$

→ law of Exp.

$b^x \cdot b^y = b^{x+y}$ $\quad b^{xy} = b^{x \cdot y}$ $\quad \frac{b^x}{b^y} = b^{x-y}$

Logarithmic function → $f(x) = \log_b x$

$b > 0$ & $b \neq 1$ Domain : $(0, \infty)$ Range : $(-\infty, \infty)$

→ law of log.

$\log_b(xy) = \log_b x + \log_b y$ $\quad \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$ $\quad \log_b(x^r) = r \log_b x$

Trigonometric function $\csc = \frac{1}{\sin}$ $\sec = \frac{1}{\cos}$ $\frac{1}{\tan} = \cot$ Sin Cos tan

Domain : $(-\infty, \infty)$ Range : $[-1, 1]$ Sin & cos

Domain : $(-\frac{\pi}{2}, \frac{\pi}{2})$ Range : $(-\infty, \infty)$ tan

$(x + 2\pi)$ $\quad (x + \pi)$ } periodic function

Sin Cos $\quad$ tan

→ 1.5

★ Inverse functions

→ uses the horizontal line test instead of the Vertical (one-to-one if true)

↳ doesn't look for the exact no. (the approximet)

→ Sin unristricted is not one-to-one, also cos & tan

→ for domain & range refer to old final exam notes

Steps to solve "find the inverse" equations

① Remove denominator

② Get rid of brackets

③ Rearange

2.2-2.3

Limit Rules
- $x \to y^-$ (From the left)
- $x \to y^+$ (From the right)
- $x \to y$ (Both)

→ if left & right limit aren't equal than it DNE

Steps to determine the limits are infinite or not
1) Substitute
2) Simplify
3) Estimate

} if one doesnt work try the other (in this order)

1- Sub. the no. that was given to you ($\neq 0$)
2- Simplify by factoring the polynomial
3- Estimate is taking a no. close to the orgonal

EX: $\lim_{x \to 1^+} \frac{x-1}{x^2+x-2}$

1) $\frac{1-1}{1^2+1-2} = \frac{0}{0}$ X

2) $\frac{x-1}{(x-1)(x+2)} = \frac{1}{x+2} = \frac{1}{3}$

3) $x=2 \quad \frac{2-1}{2^2+2-2} = \frac{1}{4}$

Steps to find the V.A $\quad f(x) = \frac{h(x)}{g(x)}$
1) find the zeros of the denominator
2) make sure that those no. don't make the numerator equal to zero

Steps to find the V.A $\quad f(x) = \ln(g(x))$
1) find the zeros that inside of the ln

→ 2.3
Squeeze theorm
it consists of Sin or Cos & you must identify their lim which is $[-1, 1]$
$-1 \leq$ question ≤ 1 → then you take the first part of the question & repeat instead of $[-1, 1]$
- first $\leq$ middle $\leq$ + first → make both firsts = to the no.

Ex: $\lim_{x \to 1^+} f(x) = \sqrt{x-1} \sin\left(\frac{4}{x-1}\right)$

$-1 \leq \sqrt{x-1} \sin\left(\frac{4}{x-1}\right) \leq 1$
$-\sqrt{x-1} \leq \sqrt{x-1} \sin\left(\frac{4}{x-1}\right) \leq \sqrt{x-1}$
$\sqrt{x-1} \Rightarrow \sqrt{1-1} = 0$

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2.5-2.6-2.7-2.8

Conditions of continuity:
1) $f(x) = a$ defined no. (no $\circ$)
2) $\lim_{x \to a^+}$ & $\lim_{x \to a^-}$ must match
3) $\lim_{x \to a} f(x) = f(a)$

Types of discontinuity:
1) Removable $f(x) =$ und
2) Infinite has an Asymptote
3) Jump

You can tell if its continuing or not if both a^- & a^+ match

→ intermediate value theorem
· $f(a) < N < f(b)$ if there's N then a no. exist in (a, b)

→ 2.6
Rules to evaluate limits of $(-\infty, \infty)$
1) if the weight of the denominator > than the numerator than it's equal to zero.
2) if the weight of the denominator < than the numerator than it doesn't have a H.A
3) if the weight of the denominator = the no. of the numerator than it has a H.A

Steps to evaluate
1) look at which part has the stronger power or if they are equal
2) isolate the terms from the rest
3) cancel as much as possible from the (x^r)

→ 2.7 & 2.8

3.1-3.2-3.3-3.4

Set formulas I must memorise :

power : $\sqrt{x} = \frac{1}{2\sqrt{x}}$

- $f(x) = b^x \rightarrow b^x \cdot \ln b$
- $f(x) = b^{f(x)} \rightarrow f'(x) \cdot b^{f(x)} \cdot \ln b$

- $f(x) = e^x \rightarrow e^x$
- $f(x) = e^{f(x)} \rightarrow f'(x) \cdot e^x$

Higher Derivatives :

- Velocity : $V(t) = S'(t)$ {first}
- Acceleration : $a(t) = V'(t) = S''(t)$ {second}
- Speed : $|V(t)|$

→ 3.2

Derivatives of functions

Product : (1st fun.) × (2nd der. fun.) + (1st der. fun) × (2nd fun.)

Quotient : $\frac{\text{denom.} \times \text{der. of numer.} - \text{numer} \times \text{der. of denom.}}{(\text{denom.})^2}$

→ 3.3

Derivatives of Trigonometric functions

- $\sin(x) = \cos(x)$
- $\cos(x) = -\sin(x)$
- $\tan(x) = \sec^2(x)$
- $\csc(x) = -\csc(x) \times \cot(x)$
- $\sec(x) = \sec(x) \cdot \tan(x)$
- $\cot(x) = -\csc^2(x)$

tangent V.S normal

$y - f(a) = m(x-a)$ V.S $y - f(a) = -\frac{1}{m}(x-a)$

→ 3.4

- $f(x) = f(g(x)) \Rightarrow g'(x) \cdot f'(g(x))$
- $f(x) = (f(x))^n \Rightarrow n(f(x))^{n-1} \cdot f'(x)$
- $f(x) = f \cdot g \cdot h \Rightarrow h'(x) \cdot g'(h(x)) \cdot f'(g(h(x)))$

1st Derivatives

Constant = 0

Power = nx^{n-1}

Exponential = $b^x \cdot \ln b$

Natural = $x \cdot e^x$

Functions

$f(x) \cdot g(x)$ = product rule

$C \cdot f(x)$ = constant multiply rule

$\frac{f(x)}{g(x)}$ = Quotient rule

$(y - y_1) = m(x - x_1) \Rightarrow$ tangent line

pythagorean

$\sin^2\theta + \cos^2\theta = 1$

$\tan^2\theta + 1 = \sec^2\theta$

$1 + \cot^2\theta = \csc^2\theta$

TRIG. LIMITS

$\lim_{\theta \to 0} \frac{\sin(a\theta)}{b\theta} = \frac{a}{b}$

$\lim_{\theta \to 0} \frac{\cos\theta - 1}{\theta} = 0$

→ 3.9

finding rates:

1) if y or x are missing use the pythagorean formula $z^2 = x^2 + y^2$

2) depending on which derivative is missing is what you will make it = to the other derivative.

-note you will be deriving $z^2 = x^2 + y^2$. It will look like $(2x)(x') + (2y)(y') = 0$ combine and move according to what you need (x') or (y'). Typically it's $y' = \frac{x(x') \cdot \$}{y}$

3) Sub the points from step 1 into step 2.

→ 3.11

Hyperbolic functions ___h (x)

- $\sinh(x) = \frac{e^x - e^{-x}}{2}$ the diff. is $-$ to $+$ - $\cosh(x) = \frac{e^x + e^{-x}}{2}$
- $\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$ diff. is the reciprocal - $\coth(x) = \frac{\cosh(x)}{\sinh(x)}$
- $\operatorname{csch}(x) = \frac{1}{\sinh(x)}$ - $\operatorname{sech}(x) = \frac{1}{\cosh(x)}$

-note that the derivatives of hyperbolic functions is the same as the regular (page 8) but you simply add an 'h'

-note when finding y' by implicit diff. if there is a $\sqrt{x+y}$ that you use the formula $\frac{1+y'}{2\sqrt{x+y}}$ but you add y' in the num.

- Differentiate means to find the f' = (no y')
- Implication by differentiation means find y' =
- Verify the following means to see if both sides are equal
- Solve by using logarithmic means to expand using 'ln' then $\frac{\$'}{\$}$

→ 3.9

The only formula I should memorize is the pythagorean formula

$z^2 = x^2 + y^2$

$\frac{dy}{dt} = y'$ $\frac{dx}{dt} = x'$ ← The same

Remember:

$x^{m/n} = \sqrt[n]{x^m}$

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4.1-4.2

Absolute, local, min. & max values

- Absolute max. is the highest point in the graph, } you add local for this case too
- Absolute min. is the lowest point in the graph.
- Local max. is the second highest in the graph
- Local min. is the second lowest in the graph.

Critical numbers: (open interval)

Step 1. find the derivitave
Step 2. make it equal to zero & simply factor to get critical no.

Critical numbers: [closed interval]

Preform steps 1 & 2.
Step 3. Once you've attained the critical no. make sure that that no. is inside the closed interval [.]
Step 4. write your final ans. that meet the requirements.
-note that the bigger no. is Abs. max & the smallest is Abs. min.

→ 4.2

Mean value theorm:

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

or

$f'(c)(b-a) = f(b)-f(a)$

1. find the derivitave of $f(x)$ to make $f'(c)$
2. Sub into formula ↑ to get critical point.
3. check if it lies inbetween the $[a, b]$.

Rolles theorm:

$f'(c) = 0$

1. find the derivitative of $f(x)$ to make it $f'(c)$
2. Sub. $[A, B]$ separately into the originanlly formula to make sure $A = B$
3. If step 2 is correct go back to step 1 to find c (make it $= 0$) check if its between $[A, B]$

*you must write 1- continuous on $[A, B]$
2- differentiable on (A, B)

Note the following:
1. when there are more than 1 abs. min & max then you can't write the value
2. You don't look at the un-shaded
3. for critical no. (often) sometimes you use the chain, product, quotation rule → 4.2

$f'(c)(b-a) = f(b) - f(a)$ is used when they say $f'(x) \geq \leq <> \geq$

4.3-4.4-4.5

Increasing & decreasing
- $f'(x) > 0$ increasing
- $f'(x) < 0$ decreasing

local max. & min. / concave up & down
- $f''(c) >$ local min. / concave up
- $f''(c) <$ local max. / concave down

First derivative test cases:
1. $f'(x) > 0 \cdot c\, f'(x) < 0$ local max.
2. $f'(x) < 0 \cdot c\, f'(x) > 0$ local min.
3. $f'(x) = c \cdot f'(x)$ none

Steps to Solve:
1. find all critical no. (x) (make it = to 0)
2. Sub no. between the critical no. into the f'(x) formula. (based on - & + it tells you if its increasing or decreasing)
3. Sub original critical no. into original form to find local max & min.

Second derivative test cases ↗ steps to solve?
1. find c by $f''(c) = 0$
2. sub c into f''(c) (or no. between for multiple c's)
3. Sub original c into f(c) to get local min. & max.

★ to get infliction point sub f''(c) into original form

Sub. c into — gives:
f(x) = local max. & min.

f'(x) = critical Point, decrease & increase

f''(x) = increase & decrease, concave up & down

$\frac{no}{0} = \pm\infty$ $\quad \frac{no}{\infty} = 0$

4.4 → you wont be told which method to use. You must automatically know when you see $\frac{0}{0}, \frac{\infty}{\infty}, 0\cdot\infty, \infty-\infty$

you will sub. c into original formula when you need the local min & max / infliction point

→ 4.4

$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty} \to \frac{f'(x)}{g'(x)}$

$\lim_{x\to a} f(x)g(x) = 0\cdot\infty \to \frac{f'(x)}{(g(x)^{-1})'} \to \frac{f'(x)}{-1(g(x)^{-2})g'(x)}$

$\lim_{x\to a} h(x) - k(x) = \infty - \infty \to \frac{f'(x)}{g'(x)}$

$\lim_{x\to a} \left(\frac{\ln(f(x))}{(g(x))^{-1}}\right)$

$\lim_{x\to a} [f(x)]^{g(x)} = 0^0, \infty^0, 1^\infty \to$ 1) $e^{\lim_{x\to a} g(x)\ln(f(x))}$ 2) $A = \lim_{x\to a} g(x)\cdot\ln(f(x))$ 3) $\lim_{x\to a} [f(x)]^{g(x)} = e^A$

→ 4.5

info needed to graph:
1. Domain ← almost always $(-\infty, \infty)$ the exception is if its a fraction. you must find the 0's of the denom.
2. Intercepts ← where you make x=0 to find y & y=0 to find x
3. Symmetry [f(-x) = f(x) Even] [f(-x) = -f(x) Odd]
4. Asymptotes [horizontal is y] [vertical is x]
5. Intervals of increase & decrease ← Sub no. between critical in f'(x)
6. Local max. & min. ← by f''(x) or f'(x) test
7. Concavity & Point of infliction
8. Sketch